

SSS 1-3 Research and Design Garden Flashcards 961-980

961. ENGINEERING PROBLEM	An engineering problem statement defines an unmet need, system context, affected...
962. STAKEHOLDER REQUIREMENT	A stakeholder requirement describes a needed function, performance level, usabil...
963. DESIGN REQUIREMENT	A design requirement translates needs into testable technical criteria that a pr...
964. CONSTRAINT	A design constraint is a non-negotiable limit involving budget, safety, space, r...
965. DESIGN VARIABLE	A design variable is a controllable parameter that can be changed during design,...
966. PERFORMANCE METRIC	A performance metric quantifies how well a design meets an objective or requirem...
967. DESIGN ALTERNATIVE	A design alternative is a distinct feasible configuration or strategy developed ...
968. CONCEPT SCREENING	Concept screening eliminates alternatives that fail essential requirements or co...
969. DECISION MATRIX	A decision matrix compares alternatives across weighted criteria using an explic...
970. TRADE-OFF ANALYSIS	Trade-off analysis examines competing objectives where improvement in one criter...
971. PROTOTYPE	A prototype is a simplified physical, computational, or functional representatio...
972. TEST PLAN	A test plan specifies performance questions, procedures, instruments, acceptance...
973. ACCEPTANCE CRITERION	An acceptance criterion is a measurable threshold that determines whether a test...
974. DESIGN ITERATION	Design iteration uses test evidence to modify a solution and retest performance.
975. TECHNICAL RISK	Technical risk combines the likelihood and consequence of failure or underperfor...
976. UNCERTAINTY	Design uncertainty arises from incomplete knowledge of loads, environment, users...
977. LIFE-CYCLE THINKING	Life-cycle thinking considers material acquisition, construction, operation, mai...
978. TECHNO-ECONOMIC METRIC	Techno-economic metrics connect technical performance with costs, benefits, life...
979. DESIGN DOCUMENTATION	Engineering documentation preserves requirements, assumptions, drawings, calcula...
980. ENGINEERING DESIGN REVIEW	A design review determines whether evidence supports progression, revision, or r...